

Jahangirnagar University
Department of Economics
M.Sc Final Examination 2022
Course No. 501 (Advanced Microeconomic Theory)
Time: 4 hours **Full Marks: 70**

Answer any 5 of the following questions
All questions are of equal value

- ✓ 1. a) What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach. 7
b) Define indifference set. Examine that if a preference relation satisfies strong monotonicity, its indifference curves must be downward sloping. 7

2. a) What are the twin definitions of convexity? In what respect are they different from strong convexity-examine graphically? 4
b) Show that, 5

Convexity of preferences $\Leftrightarrow USC(x)$ is convex $\Leftrightarrow u(.)$ is quasiconcave

- c) Prove that, if a preference relation satisfies monotonicity and continuity, then there exists a utility function $u(.)$ representing such preference relation. 5

- ✓ 3. Consider the Cobb-Douglas utility function: 14

$$u(x_1, x_2) = Kx_1^\alpha x_2^{1-\alpha} \quad \text{for some } \alpha \in (0,1) \text{ and } K > 0$$

- i) Derive the Walrasian demand functions and examine their properties.
ii) How do you get the indirect utility functions? Explain their properties and also verify Roy's identity.
iii) Solve the expenditure minimization problem and examine the characteristics of the compensated demand function.

- ✓ 4. a) Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity. 5

- b) Show that,

$$\text{Strong Monotonicity} \not\Rightarrow \text{Monotonicity} \not\Rightarrow \text{LNS} \quad 5$$

- c) Examine the following properties of preference relations: 4

- i) Homogeneity
ii) Homotheticity

- ✓ 5. a) Define production sets. Graphically explain the properties of the production set. 5

- b) What is the elasticity of substitution (EoS). Find the elasticity of substitution for the production function $q = f(K, L) = aK + bL$. What will be the EoS if the production function is of fixed proportions? 6
- c) If the production set Y is closed and has free disposal, then what properties do the profit function satisfies? 3

6. a) Distinguish between compensating variation (CV) and equivalent variation (EV). In what respect they are different from the consumer's surplus (CS) 4
- b) Illustrate that for a decrease in price of a commodity (when both goods are normal)

$$EV > CS > CV$$

- c) Consider a consumer with the utility function: $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{2}}$ 5

i) Consider a) $\{p^0, m^0\} = \{4, 4, 300\}$

b) $\{p^1, m^0\} = \{2, 4, 300\}$

Compute the indirect utilities and costs.

- ii) Calculate EV, CV and CS. Based on their relationship examine the nature of the commodities.

- ✓ 7. a) Explain the existence, uniqueness and stability of a general equilibrium. Show that i) existence of an equilibrium occurs at a positive price, ii) stability depends on the relationship between the slopes of the demand and supply curves and iii) uniqueness is related to the slope of the excess demand function. 7

- b) Graphically explain how an external shock in the product market affects the factor market and the mechanism through which equilibrium is restored in both markets. 7

8. An exchange economy consists of two people, A and B, and two goods X and Y. A's preferences are represented by the utility function $U_A = X_A Y_A$; B's preferences are represented by the utility function $U_B = X_B + Y_B$; The initial endowments are given by

$$X_A^0 = 10; Y_A^0 = 50; X_B^0 = 50; Y_B^0 = 10$$

- a) Draw the initial allocation in an Edgeworth box. Draw each person's indifference curve through the initial endowment. 3

- b) Is the initial endowment Pareto efficient? Explain why or why not? 3

- c) What is the equation of the contract curve (write your answer as Y_A as a function of X_A)? 4

- d) Is this allocation $X_A=80, X_B=120, Y_A=50, Y_B=150$ Pareto efficient? Is it in the core? 4

Department of Economics
Jahangirnagar University
Master of Science (M.Sc) in Economics Examination 2022
Course No: 502 (Advanced Macroeconomic Theory)
Full Marks: 70 Time: Four Hours

*Answer any five of the following questions.
All parts of a question must be answered consecutively.*

1. There are four variables like output (Y), capital (K), labor (L) and knowledge or "effectiveness of labor" (A). The growth rate of X refers $\dot{X}/X$; the growth rate of L is n; growth rate of A is g and capital depreciation rate is δ . [4+3+3+4]
 - a) Briefly discuss the assumptions about input, output and production function to construct Solow exogenous growth model.
 - b) Briefly describe phase diagram for k. Why it does follow the Balanced Growth Path (BGP)?
 - c) Describe how each of the following affects the break-even and actual investment:
 - i) The rate of depreciation falls.
 - ii) The rate of technological progress rises.
 - iii) Capital's share, α rises.
 - d) Show the impacts of saving ratio in the steady state solution of Solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and y, but not growth effects.
2. Consider a Solow economy that is on its balanced growth path and production function follows Cobb-Douglas production function $Y(t) = F(K(t), A(t)L(t))$. The fraction of output devoted to investment, s, is exogenous and capital depreciation rate is δ . The growth rate of L and A are respectively n and g. [3+5+3+3]
 - a) Show the impacts of changes in the saving ratio on per capita consumption. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.
 - b) Explain the long run effect of a rise in saving on output. In the vicinity of the balanced growth path, how rapidly does the economy converge to the balanced growth path?
 - c) Find the elasticity of output per unit of effective labor on the balanced growth path, y^* , with respect to the rate of population growth, n. If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $\delta = 3\%$, by about how much does a fall in n from 2 percent to 1 percent raise y^* ?
 - d) Briefly explain growth accounting and convergence as empirical applications.
3. Consider a version of the standard Ramsey Cass Koopmans model with the lifetime utility of the representative household $U_0 = \int_0^\infty u(c_t)e^{-\theta t} dt$, where the number of families are large, but they are normalized to unity. N is the supply of labour and the population growth rate is exogenous $n > 0$. Find the optimization problem of a representative family at time 0 (zero) while the constraint is given by $\dot{d}_t = r_t a_t + w_t - c_t - n a_t$, where r_t is the net rent on return capital. [3+4+7]
 - a) Find a solution to the household problem.
 - b) Define a competitive equilibrium.
 - c) Find an equation that determines the steady state equilibrium value of capital.
4. Suppose $Y_t = F(K_t, A_t L_t)$, with having constant returns to scale and the intensive form of the production function satisfying the Inada conditions. Suppose also that $A_{t+1} = (1+g)A_t$, $L_{t+1} = (1+n)L_t$ and $k_{t+1} = L_t(w_t A_t - G_t)$. Assume time is discrete rather than continuous and each individual lives for only two periods. [5+2+3+4]
 - a) Solve the individual's maximization problem where θ determines how much individuals varies in response to differences between r (real rate of return) and ρ (discount rate). Explain the role of income effect and substitution effect to determine individuals' consumption and saving.
 - b) Find an expression for k_{t+1} as a function of k_t .
 - c) Describe how each of the following affects k_{t+1} as a function of k_t :
 - (i) A rise in n.
 - (ii) A fall in ρ .
 - (iii) A permanent increase in government purchases.
 - d) Find the speed of convergence of k to k^* . How does it differ from Solow model?

5. Consider a simplified version of the models of R&D and growth model developed by P. Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D, and B is a shift parameter. Both goods producing sector and R&D producing sector use the full stock of knowledge A . [2+4+4+4]
- How does the process to construct the framework differ from Solow model?
 - Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model.
 - Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:
 - An increase in n .
 - An increase in a_K .
 - There are two cases $\beta + \theta < 1$ or $\beta + \theta = 1$ and $n = 0$;
 - What is the difference on the dynamics of the growth rates of capital and knowledge between these two cases?
 - Distinguish between semi-endogenous growth model and fully endogenous growth model.
6. Suppose firm i 's output is $Y = [\int_{i=0}^A L(i) di]^{1/\phi}$, and $L(i)$ is used to denote both the quantity of labor devoted to producing input i and the quantity of input i that goes into final-goods production. Individual's discount rate is ρ and their instantaneous utility function is logarithmic. The utility of the representative household takes the constant-relative-risk aversion (CRRA) form is $\frac{\dot{C}(t)}{C(t)} = \frac{r(t) - \rho}{\theta}$ and θ is the coefficient of the relative risk aversion. Knowledge creation, $\frac{\dot{A}(t)}{A(t)} = BL_A$. Growth rate of the wage equals the growth rate of output. Population growth is zero. [3+4+4+3]
- When output is produced by competitive firms, what will be the demand for the holder of the patent on an idea and the profit of firm?
 - Briefly discuss the assumptions to construct the model.
 - How the long-run growth will be determined by using the present value of profits from the discovery of a new idea at time t ? What will be the equilibrium value of L_A ?
 - What will be the socially optimal level of L_A ? Explain how it is different from the equilibrium level of L_A .
7. Consider the infinitely lived representative household that derives utility from both consumption (c_t) & leisure ($1 - l_t$). The production function of the representative firm is: $f(k_t, l_t, z_t) = z_t k_t^\alpha l_t^{1-\alpha}$ $z_{t+1} = \phi z_t + \epsilon_{t+1}$ [3+2+4+5]
- where, z_t is the productivity parameter and ϵ is a white noise error term.
- Construct the household's problem and using log utility derive the short-run labor supply curve. Explain the relationship among leisure, real wage, interest rate.
 - Prove that long run labor supply is independent of wage when $\phi = \frac{1}{3}$
 - Assuming $\frac{1+r}{1+\rho} = 1$ and functional form of utility of leisure is $v(h) = \frac{\sigma}{1-\sigma} (h)^{\frac{\sigma-1}{\sigma}}$, find the elasticity of labor supply.
 - Discuss Hansen's indivisible labor supply solution as an attempt to overcome the limitations of previous model.
8. The representative producer of good i has production function $Q_i = L_i$; utility depends on consumption and Labor effort where P is known as perfect information. z_i (good specific shock) and m (monetary shock) are normally distributed. Conditional expectation is a linear equation. On the other hand, For representative consumer $\dot{C}_t = \sigma(r_t - \rho) C_t$ and wicksellian interest rate $r^* \equiv \rho + \sigma^{-1} g$. According to Yun (1996) Optimal price v_t is $v_t = p_t + \alpha \int_t^\infty [(v_s - p_s) + \eta x_s] e^{-(\alpha+\rho)(s-t)} ds$ and $\kappa \equiv \alpha^2 \eta$; Where, η is sensitivity parameter. [4+2+4+4]
- Explain the role of rational expectation in determining the effectiveness of macro policy when there is imperfect information by using Lucas' supply curve.
 - Determine new Keynesian IS equation for continuous time.
 - Determine new Keynesian PC curve for continuous time using Leibniz's rule. Explain "divine coincidence" according to Blanchard and Gali.
 - Describe the Taylor rule in the canonical New Keynesian model where x (output) and π (inflation) both are jump variables.

-THE END -

Department of Economics
Jahangirnagar University
MSc Final Exam, 2022
Course No: ECON 503 (Development Economics)
Full Marks: 70; Time: 4 Hours

Answer any five of the following questions

All parts of a particular question must be answered consecutively

1. A. How does the Neocolonial Dependence Model explain the existence and continuance of underdevelopment?

B. Contrary to the claims of the dependence theorists, how does the neoclassical counterrevolutionaries argue about underdevelopment? (7+7)

2. A. What are the key arguments embraced by the Dualistic Development Thesis?

B. How does the False Paradigm Model of International Dependence Model argue against structural models of economic development? (7+7)

3. A. Does economic growth ensure gender equity? Explain your answer with empirical evidences.

B. Why do we think that 'feminization of Agriculture' in Bangladesh has taken place? (7+7)

4. A. Explain the concept of 'convergence theory'.

B. What are the statistical Pitfalls in Baumol's unconditional convergence theory? How these were addressed? (5+9)

5. A. How does the IMF conditionality measure the impact of the economic and social policies of countries receiving assistance? What are the typical objectives behind these conditions?

B. What are the potential benefits and drawbacks of IMF conditionality for the recipient countries, particularly in terms of economic stability, growth, and social welfare? (7+7)

6. A. Why did the world population grow so fast after the industrial revolution?

B. What are the key components in the nexus of population growth and economic development? Please explain why? (7+7)

7. A. How does economic growth influence gender equality? What are the impacts on gender inequality on sustainable economic development?

B. Briefly explain the characteristics of gender prevailing among the current labour force in Bangladesh (7+7)

8. A. Write short notes on the following: (7+7)

i. Comprador Group

ii. Unpaid employment

B. In terms of 'life expectancy', why some of the countries are lagging behind from the countries with almost similar economic conditions? Please explain with country experiences.

-The End-

Department of Economics
Jahangirnagar University
M. Sc. Final Examination 2022
Course: Econometrics (Econ 504)
Time: 4 Hours Full Marks: 70

- Answer any five (05) questions
- All parts of a question must be answered consecutively.

1. [4+5+5]
- A. What are the advantages of panel data over cross-section or time series data?
- B. What is meant by a fixed effects model (FEM)? Since panel data have both time and space dimensions, how does FEM allow for both dimensions?
- C. What is meant by an error components model (ECM)? How does it differ from FEM? When will ECM be appropriate? When will FEM be appropriate?
2. [4+4+6]
- A. What is a linear probability model (LPM)? What are its limitations? What are the advantages of Probit model over other models?
- B. How would you explain the Logit model? What are the remarkable features of this model?
- C. Car ownership is a binary variable. $Y=1$ if a person owns car, zero (0) otherwise.
 The model is estimated:

$$\bar{L} = -2.5423 + 0.4347 \ln \text{Income}$$

$$t = (-2.98) \quad (3.67)$$
 Where, $\bar{L}$ = Estimated logit model
- I. Interpret the estimated logit model.
- II. From this estimated model, how would you obtain the expression for the probability of car ownership?
- III. What is the probability that a person with an income of \$14,000 will own a car? And at an income level of \$20,000? What is the rate of change of probability at the income level of \$18,000?
3. [3+3+4+4]
- A. What are the distributed-lag and dynamic model? Give examples.
- B. What is the role of lags in economics? Give two economic models as examples.
- C. How Koyck has proposed an ingenious method of estimating distributed-lag models?
- D. C. W. J. Granger (1969) has proposed to test the 'causality' between the variables. Explain the procedure using an appropriate economic example.
4. [3+7+4]
- A. What is the simultaneous equation system? Explain it with an example.
- B. What is the problem(s) that may arise if we apply the OLS method in a simultaneous equation system? Show and explain your necessary derivation.
- C. Using the stock adjustment model, we obtained the following results for the period 1980-2022.
- $$\overline{\log T}_t = \text{constant} - 0.218 \log X_{1,t-1} - 0.855 \log X_{2,t-1} + 0.864 \log T_{t-1}$$
- $$(0.051) \quad (0.170) \quad (0.035)$$
- $$R^2 = 0.987$$
- where, the figures in the parentheses are the estimated standard errors.
 T = desired stock of tractors; X_1 = relative price of tractors; X_2 = interest rate
- I. What is the estimated coefficient of adjustment?
- II. What are the short- and long-run price elasticities?
- III. What are the corresponding interest elasticities?
- IV. What are the reasons for high or low rate of adjustment in the present model?
5. [2+6+6=14]
- A. Define stationary and non-stationary time series.
- B. What do you mean by Random Walk Model? Show that Random Walk Models without drift and with drift are Non-stationary Stochastic Process.
- C. Explain the Unit root test. The following results are obtained for the US CPI series from January 1947 to January 2000 for a total of 649 observations. Use the unit root test to determine the stationarity of the CPI series:
- $$\Delta \text{CPI}_t = -0.0094 + 0.00051t - 0.00066\text{CPI}_{t-1} + 0.5473 \Delta \text{CPI}_{t-1}$$
- $$t = (-0.6538) \quad (4.3431) \quad (1.5472) \quad (16.448)$$
- $$R^2 = 0.5177 \quad d = 2.1410$$
- (The 10 % critical ζ value is - 3.1317)

6. [3+3+6+2]
- What is the structural or behavioral and reduced-form equation system? Define those with examples.
 - What is identification problem? Explain the under identification and over identification problem with examples.
 - Verify the following equation whether they are identified by both identification rules or not

Equation no.	1	Y_1	Y_2	Y_3	Y_4	X_1	X_2	X_3
1	$-\beta_{10}$	1	$-\beta_{12}$	$-\beta_{13}$	0	$-\gamma_{11}$	0	0
2	$-\beta_{20}$	0	1	$-\beta_{23}$	0	$-\gamma_{21}$	$-\gamma_{22}$	0
3	$-\beta_{30}$	$-\beta_{31}$	0	1	0	$-\gamma_{31}$	$-\gamma_{32}$	0
4	$-\beta_{40}$	$-\beta_{41}$	$-\beta_{42}$	0	1	0	0	$-\gamma_{43}$

- How could you test the "simultaneity" of a model according to the Hausman specification test?

7. [4+3+4+3]
- Explain with examples the steps of the Indirect Least Square (ILS) method to estimate a simultaneous equation.
 - How would the recursive models be used to estimate the simultaneous equations? Explain with proper example.
 - What is the necessary condition for using the 2SLS? Explain its steps with example.
 - Given the results of two regression model's:

2SLS Regression Results:

OLS Regression Results:

$$\hat{Y}_{2t} = -3521.65 + 0.6230\hat{Y}_{1t}$$

$$se = (125.352) (.0214)$$

$$t = (?) \quad (?) \quad R^2 = 0.9804$$

$$\hat{Y}_{2t} = -3234.24 + 0.6210Y_{1t}$$

$$se = (135.352) (.0209)$$

$$t = (?) \quad (?) \quad R^2 = 0.9801$$

Here, Y_1 = GDP; X_1 = private investment; X_2 = government expenditure; and Y_2 = money supply
 "Since the OLS and 2SLS results are practically identical, 2SLS is meaningless." -Comment.

8. [4+2+4+2+2]
- What is integrated stochastic process? Explain the AR, MA and ARIMA models with examples.
 - What is the error correction mechanism (ECM)? What is its relationship with cointegration?
 - What is the 'unit root' problem in a time series data? Explain the steps of Augmented Dickey-Fuller (ADF) test for the unit root problem.
 - What are the limitations of the Vector Autoregressive (VAR) model?
 - Given the results involving a variable with time series data. How many lags should be chosen and why?

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-21.7346				.000065	1.71564	1.7754	1.90246
1	126.437	296.34	16	0.000	9.9e-09	-7.09581	-6.79697	-6.16168*
2	148.47	44.066	16	0.000	7.0e-09	-7.49799	-6.96009	-5.81656*
3	168.722	40.505*	16	0.001	6.3e-09*	-7.78148*	-7.0045*	-5.35274
4	193.517	49.59	16	0.000	5.0e-09	-8.36781	-7.35176	-5.19176

Department of Economics

Jahangirnagar University

M.Sc. Final Examination 2022

Course Title : Environmental Economics

ECON: 506

Marks- 70

Time- 4 Hours

Answer any Five the following Questions. All parts of a question must be answered consecutively.

1. a) What is the main concern of Utilitarianism?

b) Critically explain the concept of preference-satisfaction utilitarianism as a basis for modern welfare economics.

c) What are Utilitarianism's weaknesses? How Rawls's 'Theory of Justice' does challenges classical utilitarianism? Explain.

3+5+6

2. a) Evaluate the empirical evidence for the Environmental Kuznet's Curve (EKC) hypothesis.

b) Discuss the policy implications of the EKC including the potential for decoupling economic growth from environmental degradation.

6+8

3. a) Distinguish between flow and stock pollutants. Analyze efficient levels of flow pollutants and stock pollutants.

b) Explain the role of spatial differentiation for emission targets

c) Find the efficient solution (M^* and A^*) for uniformly mixed and short-lived stock pollutant with the following information:

$$\text{Total Benefit, } B = \begin{cases} 96M - 0.2M^2 & \text{for } 0 \leq M \leq 240 \\ 11520 & \text{for } M > 240 \end{cases}$$

$$\text{Total Damage, } D = 0.2A^2 \text{ for } A \geq 0$$

and in steady state assume that $A = 2M$

where,

M is the Emissions flow

A is Stock Pollutant

6+4+4

4. a) What are the environmental consideration for risk assessment? Explain the risk assessment process regarding environmental degradation.

b) What is risk matrix? How much environmental risk is acceptable?

c) Explain various tools for assessing/mitigating risk.

7+3+4

5. a) Define compensating variation (CV) and equivalent variation (EV) in consumer preference. Show the relation among CV, EV and Marshallian Consumer Surplus (MCS).
 b) Suppose that an individual has the utility function $U = E^{0.25} + Y^{0.75}$ where E is the some index of environmental quality and Y is income. From an initial situation where $E = 1$ and $Y = 100$, Calculate Compensating Surplus (CS) and Equivalent surplus (ES) for an increase in E to the level 2 and for a decrease in E to the level 0.5.
 c) What is the difference between direct and indirect method of environmental valuation?

6+5+3

6. a) What are some examples of ways in which climate change will affect people around the world unequally? What are the effects of this inequality?
 b) Should the world focus efforts on mitigation or adaptation? Why?
 c) What is NDC? What are the benefits of letting each country decide its own NDC? What are the limitation of this approach? Why do you think negotiators choose this approach rather than negotiating each country's responsibility as they did in the Kyoto Protocol?

6+3+5

7. a) What is the CVM method of environmental valuation? In which regards Contingent valuation method can be considered superior to all other techniques for valuing the environment?
 b) What is the Travel cost method of valuing environment? How the Travel cost method is an example of revealed preference? Explain the limitations of the Travel cost method.

7+7

8. Write short notes on: Any four
 a) Option price and option value
 b) Certainty Equivalence and Risk premium
 c) Residual risk
 d) Net zero emission.
 e) Paris Treaty
 f) Irreversibility and Uncertainty.
 g) Loss and Damage fund

3.5 x 4 = 14

Department of Economics
Jahangirnagar University
M.Sc. Final Examination, 2022

Course: ECON 507: Health Economics

Time: Four Hours

Marks:70

Answer ANY FIVE of the following questions

All parts of a particular question have to be answered consecutively
Marks on the right margin

1. [6+4+4=14]
 - A. Define Health Economics (HE). Explain HE as a separate discipline using a schematic chart.
 - B. What are the distinguishing features of health care services, and their allocative implications?
 - C. How are supply and demand tools applied in the health care sector?
2. [6+4+4=14]
 - A. What are the public instruments in health sector that impact private decisions?
 - B. How did the cost of health care services become a major challenge in the health care sector?
 - C. How would you overcome these challenges?
3. [7+7=14]
 - A. What major problems have been identified that have made health care system itself 'sick'?
 - B. Explain your opinion on the choices that must be made in health and health care related services.
4. [6+8=14]
 - A. How does the tax system influence health insurance demand?
 - B. Explain how health insurance may affect labour supply related to retirement age and job mobility.
5. [4+10=14]
 - A. What is Clinical Decision Analysis (CDA)?
 - B. Consider a patient who is 56 years old with 'heart-burn' for several years. He may have coronary risk-factors and a six-week history of squeezing chest pain. In his negative physical examination, physicians assigned 5% coronary artery stenosis (prior-probability). Sensitivity and specificity of the exercise- ECG are 60% and 90% respectively.
 - (I) Calculate the posterior probabilities using Bayesian theorem.
 - (II) Perform steps to reach a clinical decision on performing ECG test. Consider the following utility values for best to worst outcome: 1.00, 0.75, 0.60 and 0.30.
 - (III) Graphically present a sensitivity analysis based on two pre-test likelihoods.

6. [3+6+5=14]
- A. Describe approaches in obtaining utility values for health states in Cost Utility Analysis (CUA).
 - B. Critically demonstrate Rating Scale (RS), Time Trade Off (TTO) and Standard Gamble (SG) methods for assessing preferences of respondents.
 - C. Graphically explain the relationship of cardinal utility with TTO and SG method.
7. [6+8=14]
- A. 'In a perfect health insurance market, a risk-averse individual will always be willing to pay full and fair premium for health insurance'. Discuss.
 - B. Show that in the case of adverse selection: (I) pooling equilibrium is not sustainable and (II) separating equilibrium is possible only if the proportion of high-risk individuals is much lower than that of the low-risk individuals in the pool.
8. [7+7=14]
- Write Short Notes on ANY TWO of the following:
- A. Supplier Induced Demand (SID).
 - B. Euro Qol-5D.
 - C. Cost-Effective Analysis (CEA).
 - D. Measures of calculating the value of life.

☺ GOOD LUCK ☺

Time: Four Hours

Marks:70

Answer FIVE questions, taking at least two questions from each Part A and Part B

All parts of a particular question have to be answered consecutively

Marks on the right margin

PART A:

1. [4+10=14]
A. 'Over a period, the female participation rate in the labor market increases'. Justify this statement.

B. How can a labor reach his/her most efficient time allocation point about the combination of the labor working hours and the leisure hours?

2. [2+6+6=14]
A. Illustrate the investment decision rules for taking higher education.
B. Differentiate between general on-the-job training and specific on-the-training. Graphically analyze the benefits and the costs of these two trainings.
C. Discuss using diagram(s) the signaling failure and the optimal signal when education is used as a screening device.

3. [7+7=14]
A. What are the various issues in employee selection? Briefly discuss the reasons that firms pay to spend resources on employee selection.
B. How do you explain the added worker effect and the discouraged worker effect in the labor force growth during recessions? Use graph(s) if necessary.

4. [7+7=14]
A. What are the positive and the negative aspects of the wage subsidy program? Briefly explain.
B. Why did the labor market not give rise to the predicted one wage?

PART B:

5. [14]
'Rapid advances in robotics technology have led to a growing concern over the labor market impacts of robot adoption and the disparate effects it may have across different groups of workers and occupations. Recent research findings suggest that robot adoption affects employment and wages significantly. However, the impacts of robot adoption are not borne equally among different groups of workers and occupations'. Critically evaluate these statements in the light of Mohsin Javed's (2023) research findings.

6. [14]
'While the advantages of foreign language skills on the labor market are well-studied, literature on the cost of non-native learning is still not well established'. To fill this gap, how did Juliana Bernhofer and Mirco Tonin (2022) explore the impact of the language of instruction on grades. Explain.

7. [14]
What do you mean by SAHO? Why was it mandatory during the COVID period? In light of this, Ran Gu and Ling Zhong (2023), discuss the effects of SAHO on the skill requirements in vacancy posting. Discuss their findings.

8. [2+6+6=14]
A. What are the assumptions of wage determination in a competitive labor market?
B. 'Monopsonist can exploit the workers'. Explain it graphically.
C. What are the effects of a minimum wage in a monopsony market. Briefly discuss.

Department of Economics
Jahangirnagar University
Course 512, Political Economy of Globalization
Final Exam
November 24, 2024
Marks: 70, Time: 4 Hour

Answer FIVE questions.

All parts of a question have to be answered consecutively.

Marks on the right margin

1. The World Bank has been prescribing various policy conditionality to developing countries attached to its loans. Discuss the underlying **motive** as well as the **impact** of the following three policies commonly prescribed by the World Bank. [6+4+4=14]

- a. Fiscal Discipline
- b. Removal of trade barriers
- c. Privatization of state owned industries

১. বিশ্বব্যাংক উন্নয়নশীল দেশগুলোতে প্রদানকৃত ঋণের সঙ্গে বিভিন্ন অর্থনৈতিক নীতি সংযুক্ত করে দেয় (যা কন্ডিশনালিটি নামে পরিচিত)। উন্নয়নশীল দেশগুলোর উপর বিশ্বব্যাংকের নিচে উল্লেখিত তিনটি নীতির প্রভাব আলোচনা কর। বিশ্বব্যাংক কেন এই ধরনের নীতি নির্ধারণ করে তা আলোচনা কর।

- ক) রাজস্ব শৃঙ্খলা
- খ) বাণিজ্য শুল্ক প্রত্যাহার
- গ) রাষ্ট্রীয় কলকারখানার বেসরকারিকরণ

2. a. What led to the development of Modernization theory back in the 1940s?
b. Discuss four critiques of Modernization theory with examples [4+10=14]

২) ক) চল্লিশের দশকে মডার্নাইজেশন থিওরি উৎপত্তি এবং বিকাশের পেছনের ঐতিহাসিক কোন ঘটনাগুলোর ভূমিকা ছিল।
খ) মডার্নাইজেশন থিওরির চারটি সমালোচনা উদাহরণ সহ বিশ্লেষণ করো।

3. What led to the development of Dependency theory (known as *world system theory* as well) back in the 1950s?
Discuss two main arguments and two major critiques of dependency theory with examples. [2+12=14]

৩) ক) পঞ্চাশের দশকে ডিপেন্ডেন্সি থিওরির বিকাশে ঐতিহাসিক কোন ঘটনাবলী ভূমিকা রেখেছিলো?
খ) ডিপেন্ডেন্সি থিওরির ২টি যুক্তি এবং ২টি সমালোচনা উদাহরণ সহ আলোচনা করো।

4. a. How can international negotiation help to avoid a trade-war?
b. Why does GATT-WTO system generally try to prevent non-tariff intervention in trade? How does the WTO differ from the GATT? Discuss.
c. How have the roles of the IMF and the World Bank, and the relationship between these two institutions, changed as a result of foreign debt and financial crisis? [3+5+6=14]

৪) ক) আন্তর্জাতিক আলোচনার মাধ্যমে কিভাবে বাণিজ্য-যুদ্ধ এড়ানো যায়?
খ) GATT-WTO হস্তক্ষেপ কেন আন্তর্জাতিক বাণিজ্যে non-tariff intervention বাঁধা দেয়? কিভাবে WTO, GATT থেকে পৃথক?
গ) বৈদেশিক ঋণ ও আর্থিক সংকট মোকাবেলায়, আন্তর্জাতিক মুদ্রা তহবিল ও বিশ্ব ব্যাংকের ভূমিকা ও প্রতিষ্ঠান দুইটির মধ্যকার আন্তঃসম্পর্ক কিভাবে পরিবর্তিত হয়েছে?

- 5. a. Discuss your opinion on at least 2 major critiques of the export oriented development model.
- b. Discuss a specific export oriented industry in Bangladesh which has largely altered the land, labor, and environmental pattern of a particular region of the country.
- c. Can you draw a parallel incident from the colonial history of Africa, which showed a very similar pattern? What does it tell about the continuation of international economic exploitation in the developing part of the world? [4+4+6=14]

৫) ক) রপ্তানিমুখী উন্নয়ন মডেলের অন্তত দুটি সমালোচনা বিশ্লেষণ করো।
 খ) বাংলাদেশের একটি রপ্তানিমুখী শিল্প নিয়ে আলোচনা করো যা ইতিমধ্যেই দেশের একটি অঞ্চলের জমি, পরিবেশ এবং শ্রমিক বিন্যাসে ব্যাপক পরিবর্তন এনেছে।
 গ) আফ্রিকা অথবা ভারতীয় উপমহাদেশের উপনিবেশিক সময়ের প্রেক্ষাপটে একই ধরনের একটি অর্থনৈতিক ঘটনার তুলনামূলক আলোচনা করো। এই আলোচনা থেকে আন্তর্জাতিক অর্থনৈতিক শোষণ ব্যবস্থাটির পুনরাবৃত্তি উপর আলোকপাত করো।

6. What are the economic explanations for multi-national corporations? What are some of the major effects of MNC's on home and host states? Do you think the effects have been more positive or negative? [4+6+4=14]

৬) বহুজাতিক সংস্থার অর্থনৈতিক ভিত্তি কি? Home & Host States এর উপর বহুজাতিক সংস্থার মূল প্রভাব কি? তোমার মতে, এই প্রভাবগুলো ইতিবাচক নাকি নেতিবাচক?

7. a. Does free trade benefit everyone? Discuss how removal of trade barriers may inflict conflicting interest among different groups of the economy and society.

b. In recent times, the dairy farmers in Bangladesh have been seen to throw their produced milk on the street, demanding for high protection for the local dairy farming. The dairy association of Bangladesh has demanded a high tariff of as much as 50 percent on the imported milk products. What is your opinion on protectionism in the dairy industry? [7+7=14]

৭) ক) মুক্ত বাণিজ্য কি সবার জন্যে সমানভাবে উপকারী? মুক্ত বাণিজ্য বা শুল্ক প্রত্যাহার কিভাবে সমাজ ও অর্থনীতির বিভিন্ন গোষ্ঠীর উপর আলাদা ধরনের প্রভাব ফেলতে পারে, আলোচনা করো।

খ) সাম্প্রতিক কালে, বাংলাদেশের ডেইরি খামারিদের নিজেদের উৎপাদিত দুধ রাস্তায় ফেলে প্রতিবাদ করতে দেখা গেছে। তারা স্থানীয় দুগ্ধ শিল্পের জন্যে সুরক্ষামূলক নীতি দাবি করেছেন। বাংলাদেশ দুগ্ধ শিল্প সমিতি আমদানিকৃত গুঁড়া দুধের উপর ৫০ শতাংশের কাছাকাছি শুল্ক দাবি করে আসছেন। দুগ্ধ শিল্পে এই ধরনের সুরক্ষা নীতির ব্যাপারে তোমার মূল্যায়ন আলোচনা করো।

8. a. What is Mercantilism?

b. Discuss how protectionist policies have played a vital role during the initial phase of industrial development of the western world. [6+8]

৮. ক) মার্কেন্টালিজম বলতে কি বোঝায়?

খ) পশ্চিমা বিশ্বের শিল্পায়নের একেবারে প্রাথমিক পর্যায়ে প্রটেকশনিস্ট নীতি (সুরক্ষা নীতি) কিভাবে বিশেষ ভূমিকা রেখেছিলো আলোচনা করো।